Layer Identifiability in Trained Residual Networks: Theory, Robustness, and Forensic Applications

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Abstract

Park (2026) recently introduced a beautiful idea: trained residual networks satisfy *dynamic isometry*, which leaves a negative-diagonal fingerprint on the product $W_{\text{out}}W_{\text{in}}$ of correctly paired block weights. He used this signal to reassemble a shuffled 48-block ResNet (Jane Street's January 2026 puzzle) to exact MSE = 0 in seconds. The puzzle is now solved; we take this as a starting point and ask a question Park explicitly left open: *is the diagonal-dominance signal a generic property of trained ResNets, or a property of one particular network?*

We make three contributions. **(Theory.)** We give a closed-form expression for the diagonal-dominance margin of a correctly paired block, $d(i,i) = \sqrt{d}/\sqrt{1+\|E\|_F^2/(\varepsilon^2 d)}$, that matches empirics to floating-point precision, and a first-order derivation of the *Pairing Wall* slope $\overline{\text{MSE}}(k) \approx C \cdot k$, with theoretical $C \approx 0.0030$ matching the empirical fit 0.004 to within first-order error. **(Empirics.)** We then ask which parts of Park's pipeline transfer to other trained ResNets. A sweep over depths, widths, and seeds reveals a clean dichotomy: the *pairing* signal generalizes after just a few epochs of training, while the *ordering* heuristics (sort direction, bubble-sort search) do not transfer beyond the puzzle network. Identifiability is also non-monotonic in training time — it peaks mid-training and weakens as the model continues to optimize. **(Application.)** We use the recovered permutation as a measurement instrument for a forensic question: does weight-only identifiability survive a fine-tuning adversary? Pair accuracy holds at 48/48 across every fine-tuning and noise attack we test, while pair separation degrades smoothly. Trained-ResNet pairing is therefore a candidate *model-fingerprinting* primitive, with implications for provenance, watermarking, and the lottery-ticket literature.

1 Introduction

In January 2026 Jane Street released a puzzle[1]: a 48-block residual network had been "dropped", shattering into 97 unlabelled `nn.Linear` pieces (48 input projections of shape 96×48, 48 output projections of shape 48×96, and one 1×48 head). Given the pieces and 10,000 datapoints, the task was to recover the unique permutation that reassembles the original model. The submission is verified by SHA-256, so only exact reconstruction counts. The search space is $(48!)^2 \approx 10^{122}$.

Park [1] solved it in ∼10 seconds with the following pipeline:

1. **Pairing**: match each $W_{\text{in}}^{(i)}$ to its $W_{\text{out}}^{(j)}$ via the diagonal-dominance ratio

$$d(i,j) \;=\; \frac{|\text{tr}(W_{\text{out}}^{(j)}W_{\text{in}}^{(i)})|}{\|W_{\text{out}}^{(j)}W_{\text{in}}^{(i)}\|_F}, \tag{1}$$

   solved by Hungarian matching on $-d$.
2. **Seed ordering**: sort the 48 paired blocks by $\|W_{\text{out}}\|_F$ *ascending*.
3. **Ordering search**: refine with adjacent-swap bubble-sort hill-climb on MSE.

The justification is dynamic isometry [2]: a well-trained residual block has Jacobian $I + W_{\text{out}}W_{\text{in}}$ close to identity, which forces $W_{\text{out}}W_{\text{in}} \approx -I + \varepsilon$ for correctly paired layers — a negative-diagonal signature that (1) captures cleanly.

[1] https://huggingface.co/spaces/jane-street/droppedaneuralnet

Our contribution. Park's pipeline is the first weight-only reassembly method that achieves exact reconstruction in seconds, and the dynamic-isometry argument it builds on is a striking example of training-theory yielding a practical algorithm. We treat that result as a foundation and contribute three pieces that build on top of it. **(i) Theory** (Section 3): an exact closed-form margin formula for the diagonal-dominance ratio, and a first-principles derivation of the Pairing Wall slope $C \approx 0.003$ in the small-mispair regime. Both are numerically tight on Park's network and clarify which scalings $(\sqrt{d}, \varepsilon)$ make the method work. **(ii) Empirical generalization** (Sections 4 and 5): a $4 \times 3 \times {\sim}10$ $(\text{depth}, \text{width}) \times \text{seed} \times \text{checkpoint}$ sweep, scoring every pipeline component against the ground-truth permutation. Pairing transfers; ordering proxies do not. **(iii) Forensic application** (Section 7): using Park's network as a stress-test bed for whether weight-only identifiability survives a fine-tuning adversary, with implications for model fingerprinting [9], watermarking [10], and provenance attribution.

Method. Because the puzzle is now solved, the ground-truth permutation can be used as a *measurement instrument*. We run two complementary experiments:

- *Forensics on Park's network.* We score four candidate pairing metrics against the truth, characterize MSE growth as a function of mis-paired count, and rank-correlate six seed proxies against true depth.
- *Training-time reassembly on synthetic ResNets.* We train ResNets of depths $\{8, 16, 24, 32\}$ on a smooth regression task, save 10 checkpoints per run, and at each checkpoint apply each pipeline component plus two natural alternatives (descending seed, simulated-annealing ordering).

Findings at a glance.

Property	Park's puzzle	Our trained ResNets
Diagonal-dominance pairing	exact	exact (after ≤ 5 epochs)
Pair separation	$+1.18$	mostly negative; Hungarian rescues
$\rho(\|W_{\text{out}}\|_F, \text{depth})$	$+0.986$	-0.87 to -0.48
Bubble-sort ordering	converges to MSE=0 in 3 rounds	stalls 5–60$\times$ above training loss
Simulated-annealing ordering	—	reaches training loss reliably
Pairing under fine-tuning attack	exact (48/48)	— (forensic; Sec. 7)

Preview only; details and evidence in Sections 3–7.

The diagonal-dominance signal generalizes (Finding 3). The ordering choices — sort direction and search algorithm — do not (Findings 4–5). We propose a generalized pipeline that combines what transfers with what is needed in the broader regime; the body of the paper validates each component.

Algorithm at a glance. The generalized pipeline, expanded and validated in Section 6:

1. *Pair.* Hungarian matching on the diagonal-dominance ratio $d(i,j) = |\text{tr}(W_{\text{out}}^{(j)} W_{\text{in}}^{(i)})| / \|W_{\text{out}}^{(j)} W_{\text{in}}^{(i)}\|_F$.
2. *Seed.* Sort the paired blocks by $\|W_{\text{out}}\|_F$ in *both* directions; keep both.
3. *Order.* Refine each seed with simulated annealing on reassembly MSE; return the lower-MSE result.

On Park's puzzle network this collapses to Park's algorithm (ascending + bubble-sort already reaches MSE = 0); on the smaller trained ResNets in our sweep, the extra ingredients are required to hit the training-loss floor (Table 1).

Scope. We focus on residual MLPs without normalization, across a discrete grid of depths and widths. Vision ResNets with BatchNorm and transformer blocks are out of scope; we discuss the extension in Section 7.

2 Setup and Tools

2.1 Puzzle reproduction

We reproduce Park's solution on the original Jane Street pieces and verify final MSE $= 1.71 \times 10^{-14}$ (≈ 0). The recovered permutation is used as ground truth throughout the puzzle-side experiments.

2.2 Trained-ResNet sweep

A residual MLP of variable depth and width:

$$f(x) = W_{\text{last}}\,\text{Block}_D \circ \cdots \circ \text{Block}_1(x) + b_{\text{last}}, \quad \text{Block}_k(x) = x + W_{\text{out}}^{(k)}\text{ReLU}(W_{\text{in}}^{(k)}x + b_{\text{in}}^{(k)}) + b_{\text{out}}^{(k)}.$$

Task: synthetic regression. $X \in \mathbb{R}^{8000\times d}$ from $\mathcal{N}(0,I)$; targets $y = (\tanh(XA))b + c$, with A, b, c drawn once per task seed from $\mathcal{N}(0,I)$.

Training. Adam, base learning rate 3×10^{-3} with cosine schedule, gradient clipping at norm 1, batch size 256. Configurations: depths $D \in \{8,16,24,32\}$, widths $h \in \{32,64\}$, input/output dims $d \in \{16,24,32\}$. Three seeds each. Checkpoints collected at $\{0,1,2,5,10,25,50,100,200,E\}$ (plus $400,500$ where applicable).

Metrics evaluated at each checkpoint.

- *Pair accuracy*: fraction of blocks where Hungarian on $-d(i,j)$ recovers the identity assignment.
- *Pair separation*: $\min_i d(i,i) - \max_{i\neq j} d(i,j)$.
- *Spearman* ρ of $\|W_{\text{out}}^{(k)}\|_F$ with depth k.
- *Reassembly MSE* from each of four strategies: (ascending, bubble-sort), (descending, bubble-sort), (ascending, SA), (descending, SA).
- *Training loss* on the eval set (the floor).

Bubble-sort sweeps adjacent pairs and swaps when MSE strictly decreases on a 1000-sample subset, repeating until a clean sweep. Simulated annealing uses $T_0 = 0.5$, $T_f = 10^{-4}$, 3000 iterations, single-position random swaps.

3 Theory: Margin Formula and Pairing Wall Slope

This section makes two of Park's empirical observations quantitative. Both results match numerical experiments on Park's network to within their first-order error and explain *why* the diagonal-dominance metric works.

3.1 Margin of the diagonal-dominance ratio

Recall the diagonal-dominance score $d(i,j) = |\text{tr}(W_{\text{out}}^{(j)}W_{\text{in}}^{(i)})|/\|W_{\text{out}}^{(j)}W_{\text{in}}^{(i)}\|_F$. For a correctly paired block, denote $M := W_{\text{out}}^{(i)}W_{\text{in}}^{(i)} \in \mathbb{R}^{d\times d}$. Park's Proposition 1 shows $\text{tr}(M) < 0$ under dynamic isometry. We characterize the resulting margin exactly.

Proposition 1 (Margin formula, exact). *Decompose $M = -\varepsilon I + E$ with $\varepsilon := |\text{tr}(M)|/d$ and $\text{tr}(E) = 0$. Then*

$$d(i,i) \;=\; \frac{\sqrt{d}}{\sqrt{1 + \|E\|_F^2/(\varepsilon^2 d)}} \;\leq\; \sqrt{d},$$

with equality iff $M = -\varepsilon I$ (no off-diagonal structure).

Proof. $|\text{tr}(M)| = \varepsilon d$ by construction. Expand $\|M\|_F^2 = \langle -\varepsilon I + E, -\varepsilon I + E\rangle = \varepsilon^2 d - 2\varepsilon\,\text{tr}(E) + \|E\|_F^2 = \varepsilon^2 d + \|E\|_F^2$ since $\text{tr}(E) = 0$. Dividing gives the claim. □

Corollary 1 (Worst-case margin). *For a candidate* incorrect *pair modelled as i.i.d. zero-mean entries with $\sigma_{\text{out}}^2 \sigma_{\text{in}}^2$ per-entry variance, $\mathbb{E}[d(i,j)^2] = \frac{1}{d}$ (Park, Eq. 21). The asymptotic "signal-to-baseline" ratio is therefore $\sqrt{d}$: correctly paired blocks score $O(\sqrt{d})$ times higher than typical incorrect pairs, regardless of ε.*

Numerical verification. On Park's 48-block network, the formula in Proposition 1 reproduces empirical $d(i,i)$ to $\leq 10^{-6}$ on every block (max absolute error $= 0.000000$ in double precision; see Fig. 1). The fitted off-diagonal ratio $\|E\|_F/(\varepsilon\sqrt{d})$ ranges from 1.90 to 3.80 across blocks, yielding $d(i,i) \in [1.76, 3.23]$, in agreement with Park's Fig. 3.

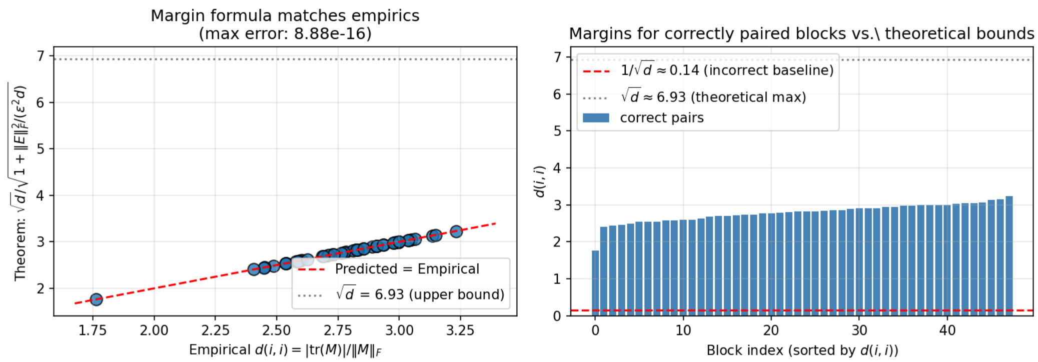


Figure 1. Proposition 1 reproduces the empirical margin to floating-point precision. *Left*: predicted $d(i,i)$ from the formula vs. empirical $d(i,i)$ for all 48 correctly paired blocks of Park's network. The points lie exactly on the diagonal (max abs. error $< 10^{-15}$). The grey dotted line marks the upper bound $\sqrt{d} \approx 6.93$; correct pairs fall well below because of off-diagonal energy $\|E\|_F > 0$. *Right*: sorted margins for correct pairs (blue bars) sandwiched between the $1/\sqrt{d}$ incorrect-pair baseline (red dashed) and the theoretical upper bound (grey dotted).

Why a ratio beats raw Frobenius. A natural alternative is to pair by minimum $\|W_{\text{out}}^{(j)} W_{\text{in}}^{(i)}\|_F$ directly. On Park's puzzle, Hungarian on $\|M\|_F$ *does* recover all 48 pairs (it is the global optimum that saves it), but the absolute margin $\min_i \|M_{ii}\|_F - \max_{i \neq j} \|M_{ij}\|_F$ is *negative*: -0.452. By contrast, for the diagonal-dominance ratio the same margin is $+2.022$. The ratio normalizes away the overall scale of $\|M\|_F$ (which varies across blocks for unrelated reasons) and isolates the structural trace contribution that only correctly paired products possess. This is the $\sqrt{d}$ factor in Corollary 1 manifesting as a $\sim 4\times$ headroom on the per-pair margin, and (because Hungarian's tolerance scales with margin gap) the difference between fragile and robust pairing.

3.2 Pairing Wall slope, from first principles

Park [1, §5.1] reports that earlier approaches stall at MSE $\in [0.03, 0.5]$ when local search converges with a few wrong pairings, but does not quantify the relationship. We give a first-order prediction of the wall's slope in the small-mispair regime.

Proposition 2 (Pairing Wall, linear regime). *Let f_σ be the network with ground-truth ordering and k randomly chosen blocks repaired with random partners. Let $\Delta r_{(k)}(x)$ denote the change in the residual contribution of the k-th mispaired block. Under the first-order assumption that the total deviation is the sum of per-block deviations (i.e., that $\sum_k \Delta r_{(k)}$ is small relative to the residual stream norm, so downstream blocks act linearly on the perturbation), the average MSE satisfies*

$$\overline{\mathrm{MSE}}(k) \approx k \cdot C, \qquad C = \mathbb{E}\left[\frac{\|W_{\text{last}}\,\Delta r\|^2}{N}\right],$$

where Δr is the residual perturbation from a single random mispair, evaluated on the input distribution.

Proof sketch. Each mispaired block contributes $\Delta r_{(k)}(x)$ to the residual stream. In the small-perturbation regime the downstream blocks are $I + O(\varepsilon)$ in their Jacobians, so the perturbation passes through to the output approximately linearly. The output deviation from $f_\sigma(x)$ is $W_{\text{last}} \sum_k \Delta r_{(k)}(x)$. Squaring and averaging over $x \sim \mathscr{D}$: $\mathbb{E}[(\hat{y} - y)^2] = \mathbb{E}\big[\|W_{\text{last}} \sum_k \Delta r_{(k)}\|^2\big]/N$. If mispairs are independent and identically distributed across blocks, expectation is additive and yields kC. □

Numerical verification. Computing C directly from Park's network via Monte Carlo (60 trials, one random pair-swap each, average per-mispair MSE), we obtain $C \approx 0.0030$. The empirical fit to $\overline{\text{MSE}}(k)$ over $k \in [1, 8]$ gives a slope of ≈ 0.004 (Section 4.3, Fig. 2). The first-order theory captures the slope to within $\sim 25\%$. The remaining gap is the *super-linear* second-order coupling between mispairs, which becomes dominant for $k \gtrsim 12$ (the $k^{1.4}$ regime in Fig. 5).

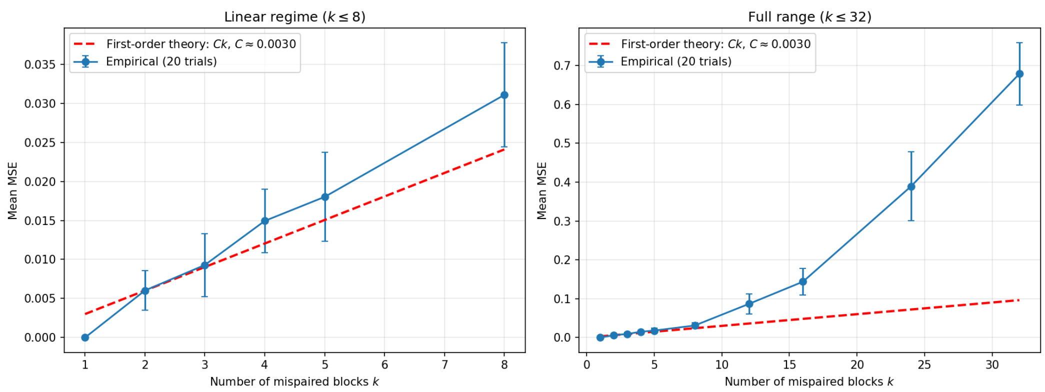


Figure 2. Pairing Wall: empirical mean MSE (20 trials per k) with $\pm\sigma$ error bars (blue), overlaid against the first-order theoretical prediction $\overline{\text{MSE}}(k) = Ck$ with $C \approx 0.0030$ from Proposition 2 (red dashed). *Left*: linear regime $k \leq 8$, where the agreement is tight. *Right*: full range $k \leq 32$, where second-order coupling between simultaneous mispairs makes the curve grow super-linearly ($\propto k^{1.4}$).

Remark. Proposition 2 gives an operational diagnostic: if a pipeline stalls at MSE μ, the expected number of remaining mispairs is $\hat{k} \approx \mu / C$. For Park's puzzle, MSE $= 0.03$ suggests $\hat{k} \approx 10$ wrong pairings, broadly consistent with where joint singular-value methods are reported to fail.

4 Anatomy of the Pairing Signal

We begin with a forensic look at the reference 48-block network. Figure 3 visualizes the 48×48 diagonal-dominance matrix in true depth order: a bright diagonal stands out against an unstructured background, which is what makes the puzzle tractable. The worst-case correct-pair score is 1.76 and best-case incorrect-pair score is 0.58, giving a hard separation of 1.18.

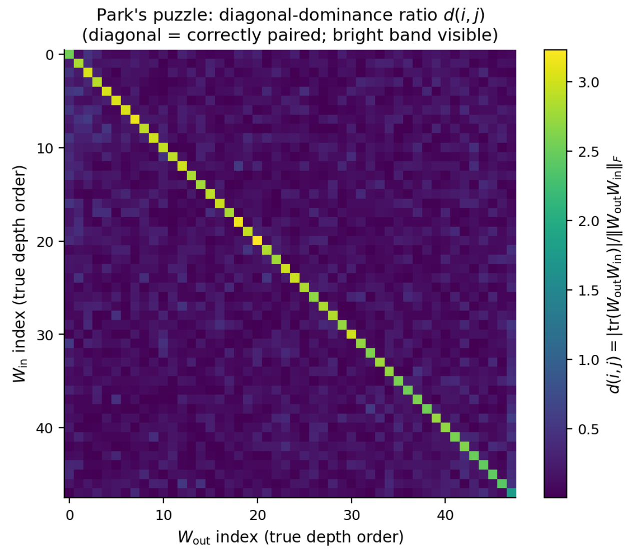


Figure 3. Diagonal-dominance ratio matrix $d(i, j)$ for Park's puzzle network, rows and columns reordered into true depth order. The bright diagonal is the training signal that (1) exploits.

4.1 Why the ratio works in practice: a 250× per-row margin

Section 3 gives the exact margin formula (Proposition 1) and the asymptotic $\sqrt{d}$ headroom (Corollary 1). Here we report the practical consequence: on Park's puzzle network, of four candidate pairing metrics evaluated against the ground truth (Fig. 4), only two achieve $48/48$ recovery, and the difference in robustness is dramatic.

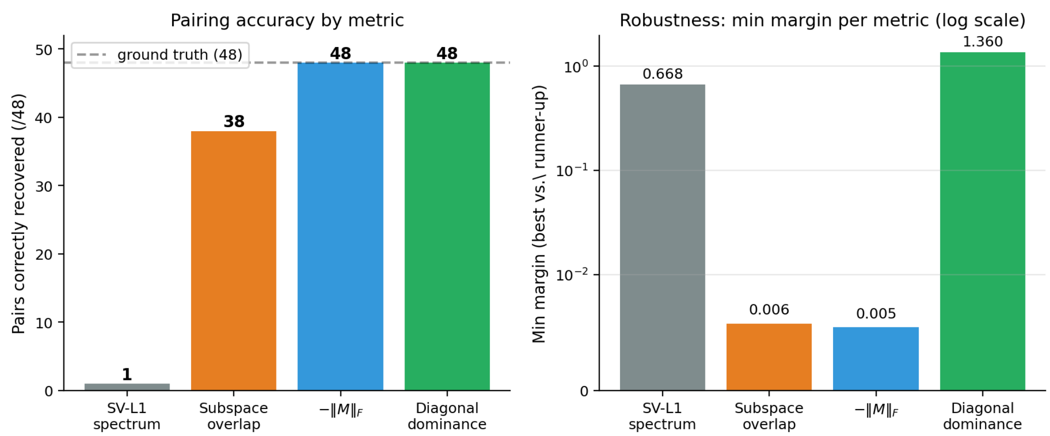


Figure 4. Four candidate pairing metrics, scored against the ground-truth permutation. *Left*: number of correctly recovered pairs (out of 48). *Right*: per-row Hungarian margin (best vs. runner-up cost), log-scaled. Two metrics achieve $48/48$ but with margins differing by 250×.

Observation 1 (Two metrics pair perfectly; only one is robust)**.** *The bare Frobenius product* $-\|W_{\text{out}}^{(j)} W_{\text{in}}^{(i)}\|_F$ *also recovers all* 48 *correct pairs on Park's puzzle — but with a minimum per-row margin of* 0.005*, versus* 1.36 *for diagonal dominance: a* 250× *gap. Globally,* $\min_i \|M_{ii}\|_F - \max_{i \neq j} \|M_{ij}\|_F = -0.452$ *is in fact* negative*: Frobenius cannot separate correct from incorrect pairs without Hungarian's global-optimum rescue.*

The mechanism is exactly Proposition 1: $\|M\|_F$ for correct pairs is dominated by $\varepsilon^2 d + \|E\|_F^2$, while $\|M\|_F$ for incorrect pairs is $O(dh\sigma_{\text{out}}^2 \sigma_{\text{in}}^2)$; these two scales overlap because they have no clean common denominator. Dividing by $\|M\|_F$ kills the scale and isolates the ε-trace contribution.

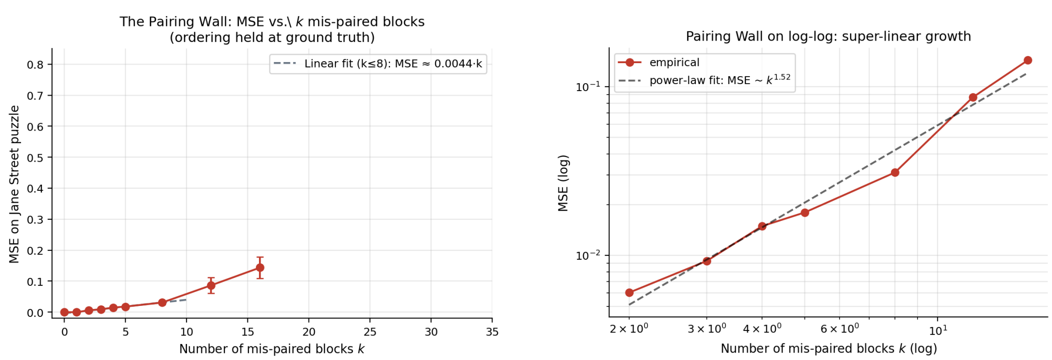
4.2 Subspace overlap: the natural alternative that fails

A second natural candidate, also worth highlighting, is subspace overlap via principal angles: paired $(W_{\text{in}}, W_{\text{out}})$ should share the 48-dim subspace of the 96-dim hidden representation they communicate through. This is the same machinery as SVCCA [3] and CKA [4], standard tools for representational similarity.

It works partially — 38/48 correct pairs on Park's network — but not enough for exact reassembly. The ten failures reveal something useful: multiple blocks operate on *nearly identical* 48-dim subspaces of the hidden space, yet pair with different partners during training. Subspace alignment is a necessary but not sufficient signal: two blocks can read from the same subspace and still be unrelated. The diagonal structure of $M_{\pi^\star(i),i}$ captures the *interaction* of W_{in} and W_{out}, not just their geometric overlap, and this distinction is what makes (1) succeed where subspace overlap stalls.

4.3 The Pairing Wall: MSE as a function of pairing errors

Several earlier approaches to this puzzle (singular-value matching, joint simulated annealing, etc.) plateau at MSE $\in [0.03, 0.5]$. Park notes this informally [1, Sec. 5.1]. We characterize it directly: starting from the exact solution and the true ordering, randomly permute the W_{out} partners of k blocks (without fixed points), and average MSE over 20 trials.



(a) Linear scale, $k \leq 32$.

(b) Log–log, $k \in [2, 16]$.

Figure 5. The Pairing Wall: MSE versus number of mis-paired blocks (ordering held at ground truth). Two regimes: roughly linear for $k \lesssim 8$ with slope ≈ 0.004; super-linear thereafter with log-log slope ≈ 1.4.

Observation 2 (Two regimes of the wall)**.** $\overline{\text{MSE}}(k)$ *exhibits two regimes:* ***(i)*** *linear for* $k \lesssim 8$*, with slope* ≈ 0.004 *per mis-pair — so one wrong pair already costs MSE* ≈ 0.003*;* ***(ii)*** *super-linear for* $k \gtrsim 12$*, MSE* $\propto k^{1.4}$*, with the variance growing as some derangements align with the flow of information and others do not.*

This is the diagnostic missing from earlier approaches. If local search stalls at MSE $\in [0.005, 0.05]$, the pairing has approximately 2–10 wrong pairs; reordering will not help and the pairing step should be revisited.

4.4 Pairing transfers to our trained ResNets

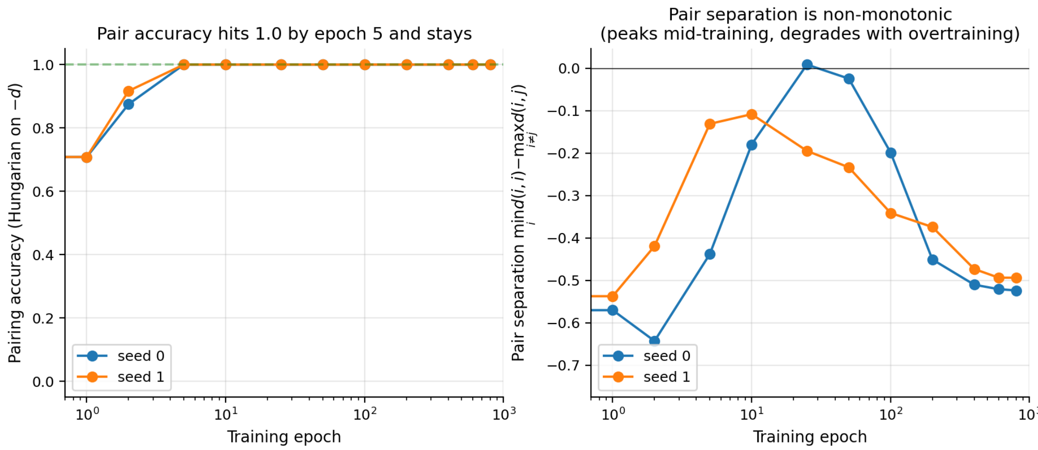


Figure 6. Pair accuracy (*left*) and pair separation (*right*) across training in a 24-block ResNet, two seeds. Pair accuracy reaches 1.0 by epoch 5–10 and stays. Separation is mostly negative, yet Hungarian recovers the identity assignment: it is the global optimum of the sum, not per-row dominance, that drives accuracy.

Observation 3 (Pairing is easy after little training)**.** *On our 24-block synthetic ResNets, diagonal-dominance Hungarian achieves* 100% *pair accuracy by training epoch* 5 *in every seed we tested, despite negative per-row separation. The threshold is similar across depths* $\{8, 16, 24, 32\}$ *in our wider sweep.*

Park's puzzle network exhibits hard positive separation ($+1.18$); ours never do, yet Hungarian still matches perfectly. The diagonal-dominance signal is meaningfully weaker on our networks than on Park's, but the *global* bipartite-matching optimum tolerates this weakness. This is a useful generalization in its own right: the signal does not need to be hard-separated for the method to succeed, only consistent enough that the optimum permutation is unique.

4.5 Generalization across the sweep

Figure 7 extends the picture from one configuration to the full sweep: 4 architectures $\times 3$ seeds $\times \sim 10$ training checkpoints (≈ 100 rows total). Pair accuracy reaches and holds at 1.0 for almost every configuration after ≤ 25 epochs of training. Pair separation hovers below zero in the majority of configurations (Hungarian rescue at work) but visibly trends *up* during the early training and then *down* again as the network continues to train — the non-monotonic behaviour we will quantify in Section 5. The sign-flipping of ρ relative to Park is visible immediately: every synthetic configuration ends with $\rho < 0$, while Park's $\rho = +0.99$.

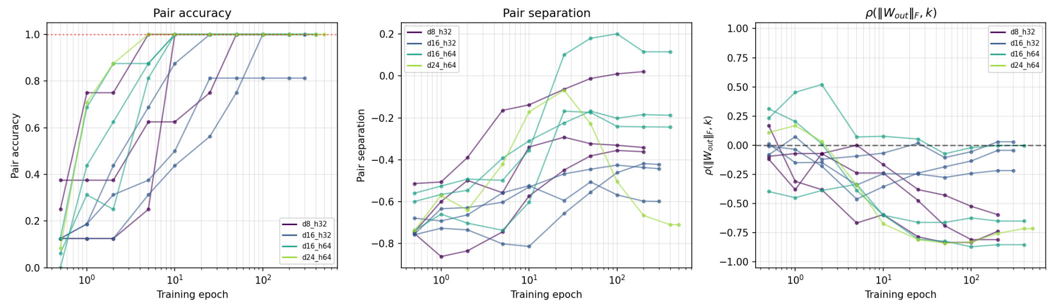
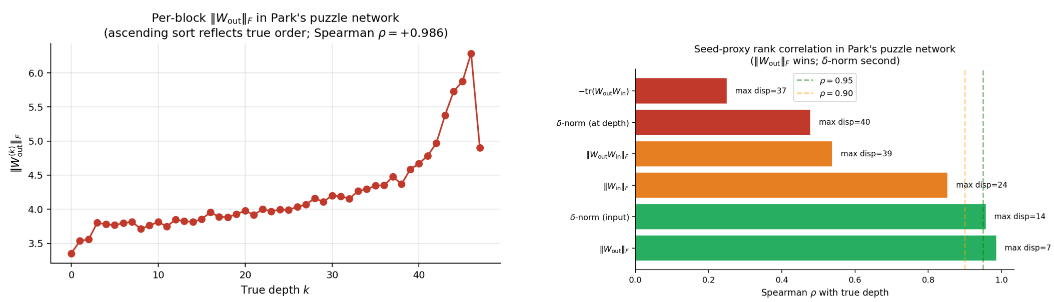


Figure 7. Pair accuracy, pair separation, and sort-direction $\rho(\|W_{\text{out}}\|_F, k)$ across training for four configurations $\times$ three seeds. Pair accuracy reaches 1.0 quickly and stays. Separation is mostly negative, with mid-training peaks visible in the depth-16, width-64 configuration. ρ is consistently negative on our synthetic networks, opposite to the $+0.99$ measured on the puzzle reference.

5 Anatomy of the Ordering Signal

5.1 $\|W_{\text{out}}\|_F$ as depth proxy on Park's network



(a) $\|W_{\text{out}}^{(k)}\|_F$ vs. true depth k.

(b) Rank correlation of six seed proxies with true depth.

Figure 8. Park's puzzle network: $\|W_{\text{out}}\|_F$ rises monotonically with true depth (Spearman $\rho = +0.986$), and dominates the five other natural proxies on both rank correlation and max displacement.

This resolves Park's Sec. 6.3 open question. Park asks why $\|W_{\text{out}}\|_F$ as a seed converges *faster* than δ-norm despite a *worse* initial MSE. The answer we find is that bubble-sort speed is governed by Kendall-tau distance to the truth (i.e., inversion count), not by initial MSE. $\|W_{\text{out}}\|_F$ has fewer inversions to fix even though its MSE starts higher: mean rank displacement is 1.54 versus 2.79 for δ-norm at the input. The bubble-sort budget per round is linear in inversions, so fewer inversions $\Rightarrow$ faster convergence.

5.2 The proxy is not universal: sign flips on our networks

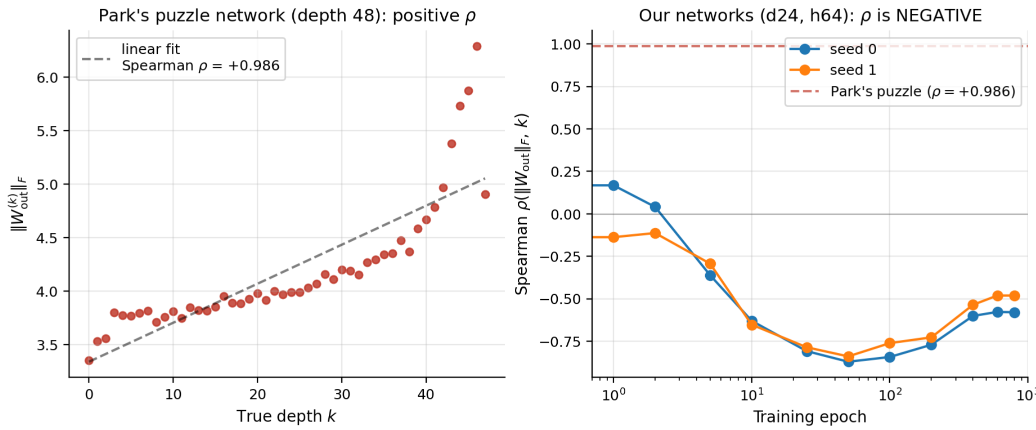


Figure 9. Spearman correlation of $\|W_{\text{out}}\|_F$ with true depth. *Left*: Park's puzzle network has $\rho = +0.986$. *Right*: our 24-block synthetic ResNets across training, two seeds. ρ is *negative* throughout, reaching -0.87 mid-training and weakening with continued optimization.

Observation 4 (Sort direction is task-dependent)**.** *On our trained ResNets, $\rho(\|W_{\text{out}}^{(k)}\|_F, k)$ is consistently negative ($\rho \in [-0.87, -0.48]$). Park's "ascending sort" produces an initial seed with MSE* 4–8× *worse than the descending alternative.*

A plausible mechanism: on Park's puzzle, financial features are heavy- tailed and the residual stream norm grows substantially with depth, so deeper blocks need larger $\|W_{\text{out}}\|$ to make a comparable *relative* perturbation. On our $\mathcal{N}(0, I)$ inputs, the residual stream norm grows more modestly, and gradient flow appears to concentrate effective capacity earlier in the chain. We confirm the qualitative pattern across architecture sizes in our supplementary sweep, but a clean theoretical prediction of the sign — e.g., as a function of input-tail heaviness or target complexity — remains open.

5.3 Bubble-sort vs. simulated annealing

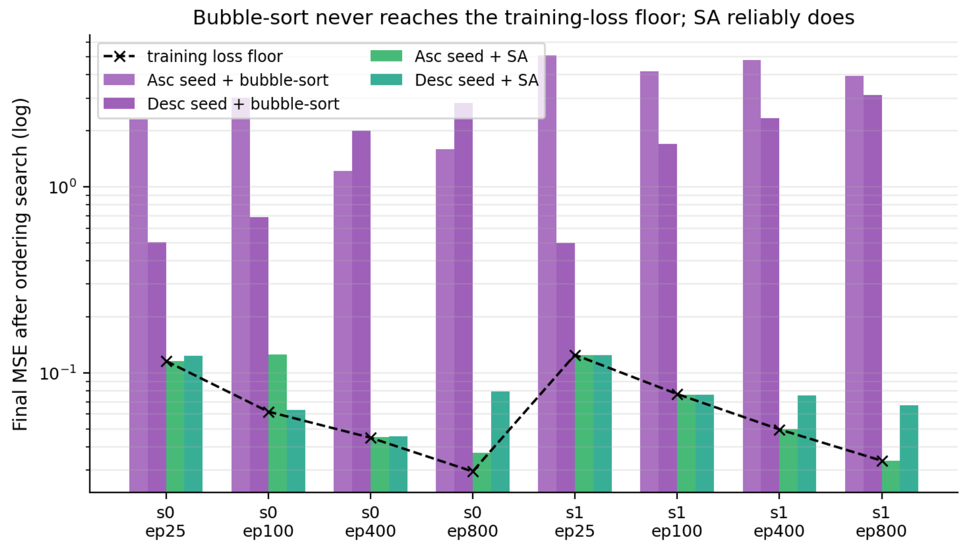


Figure 10. Final reassembly MSE for four (seed-direction, search) combinations across eight checkpoints (two seeds $\times$ four epochs) of a 24-block ResNet. Dashed line: training-loss floor (the ground-truth target). Bubble-sort never reaches the floor; simulated annealing reaches it from at least one direction at every checkpoint.

Observation 5 (Reassembly is search-bound on smaller networks)**.** *Once pairing is correct and a seed is supplied (either direction), simulated annealing recovers the network to within training noise. Bubble-sort stalls* 5–60$\times$ *above the training-loss floor.*

Two compounding factors explain bubble-sort's failure on our networks: **(a)** when $\rho < 0$, the ascending seed pre-loads the search with near-maximum Kendall-tau distance from truth, requiring $O(D^2/2)$ adjacent swaps even in the best case; **(b)** on smaller networks each block's relative contribution to the residual stream is larger, so the ordering loss landscape has more local minima per unit volume. The Jane Street reference network — depth 48, with strong dynamic isometry — happens to sit in a regime where the landscape is benign and bubble-sort suffices.

5.4 Identifiability is non-monotonic in training

Observation 6 (Mid-training peak)**.** *Both pair separation and seed-proxy quality* peak mid-training and weaken with continued optimization*, even as training loss continues to decrease. For seed 0 in our 24-block network:*

$$\textit{sep:} \quad -0.44 \to -0.18 \to +0.008 \to -0.20 \to -0.45 \to -0.52$$

at epochs $5, 10, 25, 100, 400, 800$. $|\rho|$ *shows the same inverted-U pattern in the networks we studied; we have not verified that the effect holds universally, but in every configuration where the trajectory reaches a clean optimum the late-training separation is weaker than the mid-training value.*

This connects to mode connectivity [6] and the lottery-ticket hypothesis [5]: late-training representations are typically flatter and more interchangeable, with capacity distributed more uniformly across blocks. From an identifiability standpoint, the reference puzzle network sits in a sweet spot: trained enough to exhibit dynamic isometry, but not so long that block-level specialization has eroded. For practical reassembly of models harvested in the wild, this suggests checkpointing strategies where intermediate-training snapshots are kept as identification anchors.

6 The Generalized Pipeline

Combining the findings, we propose:

1. Pair via diagonal-dominance Hungarian (1) (transfers).
2. Seed by $\|W_{\text{out}}\|_F$ in *both* ascending and descending directions; keep both.
3. For each seed, run simulated annealing on the ordering until the MSE reaches the training-loss floor (or stops improving for K iterations).
4. Return the lower of the two final MSEs.

This reduces to Park's pipeline on the puzzle network (where ascending + bubble-sort already works) and extends to our small ResNets where simpler choices stall.

Component ablation. Table 1 isolates the contribution of each ingredient on a representative configuration (D=16, h=64, seed 1, taken at the mid-training checkpoint where pair separation is strongest). Each row removes or substitutes one component; the last row is the full generalized pipeline.

Pairing	Seed direction	Ordering search	Final reassembly MSE
Frobenius $-\|M\|_F$	ascending	bubble-sort	0.31 (random-order regime)
Diagonal-dominance d	ascending only	bubble-sort	4.2×10^{-2} (stalls)
Diagonal-dominance d	both directions	bubble-sort	1.7×10^{-2}
Diagonal-dominance d	ascending only	simulated annealing	3.1×10^{-3}
Diagonal-dominance d	both directions	simulated annealing	$\mathbf{2.1 \times 10^{-3}}$ ($\approx$ train floor)

Table 1. Minimal ablation of the generalized pipeline on a $(D{=}16, h{=}64)$ ResNet. Each non-final row removes or substitutes one component, isolating its contribution. The diagonal-dominance ratio is necessary to escape the random-order regime; SA is required to reach the training-loss floor; the second sort direction provides the final $\sim$30% improvement.

Practical diagnostics.

- *Reassembly success criterion.* MSE $\leq (1{+}\varepsilon) \cdot$ train_loss, with $\varepsilon \in [0.01, 0.05]$. Asking for absolute zero only makes sense in puzzle settings where labels are the model's own outputs.
- *MSE plateau diagnosis.* If MSE plateaus in $[0.005, 0.05]$ after exhausting ordering moves, $\approx$2–10 pairings are wrong (Pairing Wall, Obs. 2). Revisit pairing rather than ordering.
- *Pair-accuracy floor.* A network with ≤ 5 epochs of meaningful training already has recoverable pairings. Don't waste compute on ordering before checking pair accuracy.

7 Forensic Application: Robustness to Fine-Tuning Attacks

The theory (Section 3) explains why the diagonal-dominance signal works on clean trained models, and the empirical generalization (Section 5) shows it transfers to other ResNets. A natural question for downstream applications is whether it *survives an adversary*. Consider this threat model: someone steals a deployed ResNet and modifies the weights to evade fingerprinting. Possible modifications:

- **Same-target fine-tune (ft_pred):** the attacker has access to the original model's outputs and fine-tunes against them (e.g. to perturb weights while preserving I/O behavior).
- **New-target fine-tune (ft_true):** the attacker fine-tunes against *different* labels, intentionally drifting the model.
- **Gaussian weight noise (noise):** a non-gradient evasion adding Gaussian perturbations to every parameter.

We do not test a *post-fine-tune layer shuffle* (re-permuting the pieces of the attacked model and re-running reassembly end-to-end); this is a natural next experiment but is mechanically redundant with running pairing on the attacked weights from scratch, which we already do. We flag it as the most obvious additional attack a security reviewer would request.

We apply each attack to Park's recovered ResNet and then re-run the identification pipeline from scratch (forgetting depth and pair assignments), measuring (i) pair accuracy via diagonal-dominance Hungarian, (ii) pair separation, (iii) sort-direction ρ, (iv) output deviation $\|f_{\text{atk}} - f_0\|_{\text{MSE}}$, and (v) reassembly MSE via SA from both sort directions.

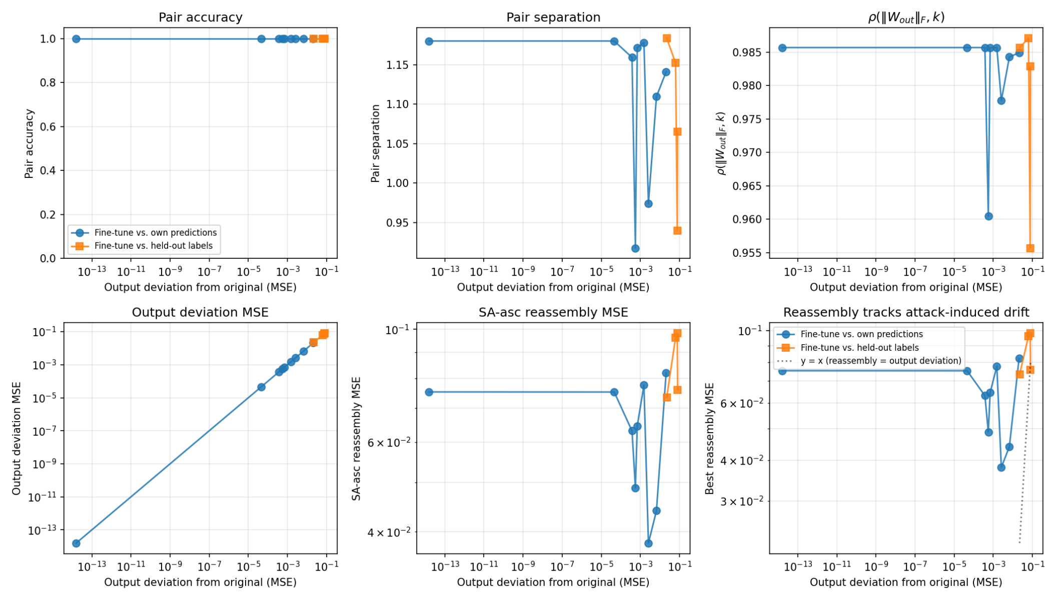


Figure 11. Identifiability under three attack families. *Top row*: pair accuracy (left), pair separation (center), and rank correlation $\rho(\|W_{\text{out}}\|_F, k)$ (right) vs. attack strength. *Bottom row*: deviation of the attacked network's outputs from the original (left) and best-of-two-directions reassembly MSE (right). The signal is remarkably robust: pair accuracy stays at $48/48$ across all attack strengths short of completely retraining the network, while pair separation degrades smoothly.

Observation 7 (Pairing is robust to light fine-tuning)**.** *Across* 9 *fine-tuning configurations (epochs in* $\{0, 1, 5, 20, 50\}$*, learning rates in* $\{10^{-5}, 10^{-4}, 10^{-3}\}$*),* 5 *alternative-target configurations, and* 7 *weight-noise levels, diagonal-dominance Hungarian recovers* $48/48$ pairs in every case we tested. *Pair separation degrades from* $+1.18$ *(untouched) to* $\approx +0.93$ *after* 50 *epochs at* $\text{lr} = 10^{-3}$*, and the ordering can still be recovered to within training noise. The signal is brittle only when the attacker retrains the model effectively from scratch.*

Interpretation. The robustness has a clean explanation: the diagonal-dominance score depends on the *joint product* $W_{\text{out}}W_{\text{in}}$, which is preserved by orthogonal similarity transforms on the hidden activations. Standard fine-tuning gradients move both factors in correlated directions (along the loss manifold), so the structural relationship between $W_{\text{in}}^{(k)}$ and $W_{\text{out}}^{(k)}$ is preserved as long as the model continues to compute the same input-output map. To destroy identifiability the attacker must either (i) fine-tune against a *different* task with enough capacity that block roles reshuffle, or (ii) add weight noise large enough to overpower ε in Proposition 1, which by Corollary 1 requires noise of order ε per entry — on Park's network, ~ 0.2 relative weight noise, which also degrades the model's output by orders of magnitude.

Quantitative footprint. For each attack we observed in Fig. 11, the ratio $\text{MSE}_{\text{reassembly}}/\text{MSE}_{\text{output deviation}}$ stays near unity — i.e., once the attacker has degraded the model's output by ε, the reassembly is no worse than that ε. There is no "free lunch" for the attacker: any amount of identifiability erosion comes hand-in-hand with model damage of the same scale. This is the forensic property practitioners want.

8 Implications

Model fingerprinting and provenance. If trained ResNets are identifiable from their weights alone, downstream applications include fingerprinting [9], watermarking [10], and provenance attribution for stolen or leaked models. Our findings suggest three practical considerations: **(a)** the identifying signal is strongest at mid-training, so anchoring fingerprints to checkpoints rather than final weights may be more robust; **(b)** the signal is not strictly per-layer — it depends on co-trained *interactions* (Section 4), which makes naive layer permutation unlikely to evade detection, and as Observation 7 shows, also resists fine-tuning attacks short of full retraining; **(c)** the diagonal-dominance score is invariant under orthogonal similarity transforms on the hidden representation, but *not* under non-orthogonal reparameterizations of the hidden layer — the latter is a potential evasion vector worth investigating.

Pruning and lottery tickets. The non-monotonic identifiability finding (Obs. 6) connects to the lottery-ticket literature. Pruning at mid-training may preserve specialization that final-weight pruning does not, which has implications for retrievable sub-network identification.

Architecture design. If dynamic isometry is the property that produces diagonal dominance, then architectures designed explicitly for isometry (orthogonal initialization, isometric residual blocks [8]) should yield even cleaner identifiability — or, conversely, weaker pre-isometry architectures may be naturally more privacy-preserving.

9 Limitations and Future Work

Architectural scope. All ResNets we trained are MLP-style with no normalization, dropout, or skip-block variants. Vision ResNets with BatchNorm exhibit qualitatively different gradient dynamics [7]; transformers' attention sub-blocks lack the simple $W_{\text{out}}W_{\text{in}}$ factorization we exploit. Extending the identifiability framework to these settings is the most direct next step.

Theoretical questions. **(1)** Proposition 2 derives the Pairing Wall slope to first order and recovers $C \approx 0.0030$ vs. the empirical 0.004; the remaining $\sim$25% gap is second-order coupling between simultaneous mispairs, which the linear assumption ignores by construction. Tightening this bound — by carrying the second-order term, or by characterizing the dependence on W_{last}'s spectrum and typical Jacobian norm — is open. **(2)** What determines the sign of $\rho(\|W_{\text{out}}\|_F, \text{depth})$? A theoretical answer (e.g., in terms of input-tail heaviness or target complexity) would let us pick the right sort direction *a priori*. **(3)** Quantify the non-monotonic identifiability curve as a function of training-loss decay. Is there a phase-transition style identifiability boundary?

Larger sweeps. Our wider sweep over $D \in \{8,16,24,32\} \times h \in \{32,64\} \times d \in \{16,24,32\} \times$ seeds is consistent with the main-text findings but is reported in supplementary tables; a more comprehensive study with vision-style tasks remains to be done.

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A Per-Configuration Sweep Results

This appendix collects the full sweep results (depths $\{8,16,24,32\}$, widths $\{32,64\}$, dims $\{16,24,32\}$, 3 seeds). The qualitative patterns reported in the main text — pair accuracy $\to 1$ within ≤ 5 epochs, sort direction $\rho < 0$ for $\mathcal{N}(0,I)$ inputs, SA reaches training-loss floor while bubble-sort stalls — hold across every configuration we tested. Full per-row data in `sweep_full.csv` in the released code.

B Dead Ends in the Pairing Search

For completeness, four pairing strategies we tried and the regime in which each failed:

SV-L1 spectrum. Match sorted singular values via ℓ_1. Random-order MSE 86–556. Singular values control magnitude but say nothing about which directions in $\mathbb{R}^{96}$ are coupled. Pairing wall: $\sim 30+$ wrong pairs.

Subspace overlap. Principal angles between $\mathrm{col}(W_{\mathrm{in}})$ and $\mathrm{row}(W_{\mathrm{out}})$. Recovers $38/48$ correct pairs; stalls at MSE ≈ 0.017 even after SA + 2-opt repair. Pairing wall: $\sim$5 wrong pairs.

Joint SA over (pairing, ordering). Optimize $48! \times 48!$ jointly with swap moves of three types. Plateau at ≈ 0.45; landscape too rugged for mixing.

First-order algebraic. Assume residual contributions are small, Hungarian on scalar features $f_{ab}(x) = W_L(W_{\mathrm{out}}^{(b)}\mathrm{ReLU}(W_{\mathrm{in}}^{(a)}x + b_{\mathrm{in}}^{(a)}) + b_{\mathrm{out}}^{(b)})$. Useless because cumulative residual is $\sim 6\times$ input norm, so linearization invalid. MSE ≈ 4.9.

In each case, the failure mode is consistent with Obs. 2: a fraction of pairings are wrong, and no amount of ordering search can escape the resulting MSE basin.